

Lab Exercise

CCP6224 Object-Oriented Analysis And Design

Total Marks: 25% (will be scaled to 40% mathematically, since lab exercise carry 40% in the syllabus)

Due Date: 29 June 2026, 11.59pm (there will not be any extension of due date)

Instructions:

1. This is a group work (a group of 3 to 4 students from the **SAME TUTORIAL SECTION**) and you must only work with among your group members. Working with anyone outside your group is considered cheating and will get zero mark. Copying from any other sources will also get zero mark. If you have trouble finding group members, please inform tutor by 15 May 2026. Individual work will get zero mark as this is a group exercise.
2. You will be given zero(0) marks if you do not submit on time. You are given ample time to submit earlier. Do not give excuses like internet problems etc.
3. You need to submit based on the submission requirements given. Submission will be online and will be informed later by your tutor.
4. Use only Java Swing to do the exercise. **DO** not use any other framework.
5. Follow the submission instruction given by your tutor and do submission before the deadline.
6. For presentation video, make the content order of presentation following the order of assessment rubric.

1. INTRODUCTION

This group assignment extends prior Java Swing and OOAD knowledge into a **real-world system** — a **Hospital Management System (HMS)**.

Students are required to design and implement a **GUI-based system using Java Swing** that manages:

- User Authentication (Login)
- Appointment Booking
- Patient Record Management
- Doctor Management

- Basic Reporting

The system must demonstrate **strong Object-Oriented Design**, proper **UML modeling**, and **functional GUI implementation**.

This assignment improves upon previous lab structure by:

- Expanding from single-feature apps → **multi-module system**
- Emphasizing **real-world OO modeling**
- Structuring **video presentation based on rubric**

2. GROUP REQUIREMENTS

- Group of **3–4 students**
- Strictly **group submission only**
- All members must appear in the **presentation video**
- Evaluation is **group-based (no individual marks)**

3. SUBMISSION REQUIREMENTS

Submit **ONE ZIP file**:

StudentID-Name.zip

Must include:

1. ☒ UML Diagrams
 - Class Diagram
 - Sequence Diagrams (for each module)
2. ☒ Java Swing Source Code
 - .java files
 - Organized into packages
3. ☒ Presentation Video (10–15 minutes)

4. SYSTEM REQUIREMENTS – HOSPITAL MANAGEMENT SYSTEM

Module 1: User Login System

- Login interface (Username & Password)
- Role-based access:
 - Admin
 - Doctor

- Receptionist
- Input validation

Module 2: Appointment Booking

- Create appointment
- Select:
 - Patient
 - Doctor
 - Date & Time
- View appointment list
- Prevent duplicate time slots

Module 3: Patient Record Management

- Add new patient
- Update patient details
- View patient history
- Store:
 - Name, Age, Gender
 - Medical history

Module 4: Doctor Management

- Add/View doctors
- Assign specialization
- Link doctors to appointments

Module 5: Basic Reporting

- Display:
 - Total patients
 - Total appointments
 - Doctor schedules

5. TECHNICAL REQUIREMENTS

- Use **Java Swing ONLY**
- Apply:
 - Encapsulation
 - Inheritance
 - Polymorphism
 - Abstraction
- Use:
 - Event handling (ActionListener, MouseListener)
 - Proper class structure
- Follow **clean architecture**

6. UML REQUIREMENTS

Class Diagram **MUST** include:

- Classes for:
 - Patient
 - Doctor
 - Appointment
 - User
 - System Controller
- Relationships:
 - Association
 - Aggregation/Composition
 - Inheritance (if applicable)

Sequence Diagrams MUST include:

1. Login Process
2. Book Appointment
3. Add Patient
4. Generate Report

7. VIDEO PRESENTATION STRUCTURE (IMPORTANT)

Must follow EXACT sequence for easy marking:

Section 1: Introduction (1 min)

- Group members
- System overview

Section 2: Class Diagram (3 mins)

- Explain design
- Show relationships
- Justify structure

Section 3: Sequence Diagrams (3 mins)

- Explain flow of:
 - Login
 - Appointment
 - Record management

Section 4: System Demonstration (5–6 mins)

Show working system:

- Login
- Add patient
- Book appointment
- View report

Section 5: Code Explanation (2 mins)

- Show:
 - OOP concepts used

- Key classes

8. MARKING RUBRIC

Criteria	Poor (1)	Below Avg (2)	Average (3)	Above Avg (4)	Good (5)
Class Diagram	Design meaningless or mostly wrong	Incomplete or wrong UML	Some UML notations missing or unclear	Mostly correct and meaningful	Complete, correct, and realistic design
Sequence Diagram	Missing or incoherent	Partially mapped	Moderately coherent	Logical and coherent	Fully coherent and detailed
Implementation	Failure or incomplete	Partial functionality	Core modules functional	Most features integrated	All modules fully functional
Design Coherence	UML and code unrelated	Partially consistent	Mostly consistent	Consistent and logical	Fully consistent and future-proof
Presentation	Missing or weak	Limited explanation	Acceptable explanation	Clear and confident	Excellent, professional delivery

9. IMPORTANT RULES

- Late submission → **0 marks**
 - Copying → **0 marks**
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